

Clinical Risk Assessment Report: Patient 56

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.51x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which reduced the patient's absolute risk by 1.9%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.78 [Top 3%] (-1.9%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Patient Age (Age) **Value: 75.0 [Top 11%] (-1.2%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.25 [Avg (54th %ile)] (-0.8%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.38 [Top 3%] (-0.7%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -3.45 [Bottom <1%] (-0.6%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -1.31 [Bottom 10%] (-0.4%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: -2.93 [Bottom <1%] (-0.4%)**
Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.1 [Top 19%] (-0.2%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Machine Learning Feature Attribution (SHAP Decision Path)

